KAIZEN MATHS

Lesson starter

Linear Algebra: Systems, Span, Basis and Transformations

Span, Independence and Basis

Name

S1

Express the given vector as a linear combination of the two vectors.

Express $\begin{bmatrix}11\\8\end{bmatrix}$ as a linear combination of $\begin{bmatrix}4\\2\end{bmatrix}$ and $\begin{bmatrix}1\\-2\end{bmatrix}$.

S2

Decide whether the vector belongs to the span of the given vectors.

Decide whether $\begin{bmatrix}17\\8\end{bmatrix}$ is in span$\{\begin{bmatrix}-4\\-2\end{bmatrix}, \begin{bmatrix}-8\\-4\end{bmatrix}\}$.

S3

Determine whether the three vectors are linearly independent.

Determine whether $\begin{bmatrix}1\\0\\-1\end{bmatrix}$, $\begin{bmatrix}0\\1\\-2\end{bmatrix}$, and $\begin{bmatrix}0\\0\\1\end{bmatrix}$ are linearly independent.

S4

Decide whether the two vectors form a basis for R2.

Do $\begin{bmatrix}-4\\-3\end{bmatrix}$ and $\begin{bmatrix}-2\\4\end{bmatrix}$ form a basis for R^2?

Show your working

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Express $\begin{bmatrix}11\\8\end{bmatrix}$ as a linear combination of $\begin{bmatrix}4\\2\end{bmatrix}$ and $\begin{bmatrix}1\\-2\end{bmatrix}$.

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Decide whether the vector belongs to the span of the given vectors.

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S3

Determine whether the three vectors are linearly independent.

Determine whether $\begin{bmatrix}1\\0\\-1\end{bmatrix}$, $\begin{bmatrix}0\\1\\-2\end{bmatrix}$, and $\begin{bmatrix}0\\0\\1\end{bmatrix}$ are linearly independent.

S4

Decide whether the two vectors form a basis for R2.

Do $\begin{bmatrix}-4\\-3\end{bmatrix}$ and $\begin{bmatrix}-2\\4\end{bmatrix}$ form a basis for R^2?

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KAIZEN MATHS

Exit ticket

Linear Algebra: Systems, Span, Basis and Transformations

Span, Independence and Basis

Name

EA1

Express the given vector as a linear combination of the two vectors.

Express $\begin{bmatrix} -20 \\ -8 \end{bmatrix}$ as a linear combination of $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -3 \\ -3 \end{bmatrix}$.

EA2

Decide whether the vector belongs to the span of the given vectors.

Decide whether $\begin{bmatrix} -8 \\ -3 \end{bmatrix}$ is in span$\{\begin{bmatrix} -3 \\ -1 \end{bmatrix}, \begin{bmatrix} -9 \\ -3 \end{bmatrix}\}$.

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KAIZEN MATHS

Exit ticket

Linear Algebra: Systems, Span, Basis and Transformations

Span, Independence and Basis

Name

EB1

Determine whether the three vectors are linearly independent.

Determine whether $\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, and $\begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}$ are linearly independent.

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KAIZEN MATHS

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Linear Algebra: Systems, Span, Basis and Transformations

Span, Independence and Basis

Name

EA1

Express the given vector as a linear combination of the two vectors.

Express $\begin{bmatrix} -20 \\ -8 \end{bmatrix}$ as a linear combination of $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -3 \\ -3 \end{bmatrix}$.

EA2

Decide whether the vector belongs to the span of the given vectors.

Decide whether $\begin{bmatrix} -8 \\ -3 \end{bmatrix}$ is in span$\{\begin{bmatrix} -3 \\ -1 \end{bmatrix}, \begin{bmatrix} -9 \\ -3 \end{bmatrix}\}$.

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Exit ticket

Linear Algebra: Systems, Span, Basis and Transformations

Span, Independence and Basis

Name

EB1

Determine whether the three vectors are linearly independent.

Determine whether $\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, and $\begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}$ are linearly independent.

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Teacher answers & working

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S1

Express the given vector as a linear combination of the two vectors.

Express $\begin{bmatrix}11\\8\end{bmatrix}$ as a linear combination of $\begin{bmatrix}4\\2\end{bmatrix}$ and $\begin{bmatrix}1\\-2\end{bmatrix}$.

Answer:

$\begin{bmatrix}11\\8\end{bmatrix} = 3\begin{bmatrix}4\\2\end{bmatrix} - 1\begin{bmatrix}1\\-2\end{bmatrix}$

Working:

Let $\begin{bmatrix}11\\8\end{bmatrix} = a\begin{bmatrix}4\\2\end{bmatrix} + b\begin{bmatrix}1\\-2\end{bmatrix}$.

Equating components gives 4a + b = 11 and 2a − 2b = 8.

Solving the two simultaneous equations gives a = 3 and b = −1.

$\begin{bmatrix}11\\8\end{bmatrix} = 3\begin{bmatrix}4\\2\end{bmatrix} - 1\begin{bmatrix}1\\-2\end{bmatrix}$.

S2

Decide whether the vector belongs to the span of the given vectors.

Decide whether $\begin{bmatrix}17\\8\end{bmatrix}$ is in span$\{\begin{bmatrix}-4\\-2\end{bmatrix}, \begin{bmatrix}-8\\-4\end{bmatrix}\}$.

Answer:

No, the vector is not in the span.

Working:

$\begin{bmatrix}-8\\-4\end{bmatrix} = 2\begin{bmatrix}-4\\-2\end{bmatrix}$, so the span is the line containing scalar multiples of $\begin{bmatrix}-4\\-2\end{bmatrix}$.

For $\begin{bmatrix}17\\8\end{bmatrix}$ to be in the span, the component ratios must match.

The ratios do not match, so no scalar t gives $\begin{bmatrix}17\\8\end{bmatrix} = t\begin{bmatrix}-4\\-2\end{bmatrix}$.

Therefore $\begin{bmatrix}17\\8\end{bmatrix}$ is not in the span.

Teacher answers & working

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S3

Determine whether the three vectors are linearly independent.

Determine whether $\begin{bmatrix}1\\0\\-1\end{bmatrix}$, $\begin{bmatrix}0\\1\\-2\end{bmatrix}$, and $\begin{bmatrix}0\\0\\1\end{bmatrix}$ are linearly independent.

Answer:

Linearly independent.

Working:

Put the vectors as columns of a matrix.

$$\begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\-1 & -2 & 1\end{bmatrix}$$

This matrix is triangular after reading the first two rows, and the third pivot is 1.

There is a pivot in each column, so the only solution to $a\begin{bmatrix}1\\0\\-1\end{bmatrix} + b\begin{bmatrix}0\\1\\-2\end{bmatrix} + c\begin{bmatrix}0\\0\\1\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$ is a = b = c = 0.

Therefore the vectors are linearly independent.

S4

Decide whether the two vectors form a basis for R2.

Do $\begin{bmatrix}-4\\-3\end{bmatrix}$ and $\begin{bmatrix}-2\\4\end{bmatrix}$ form a basis for R^2?

Answer:

Yes, they form a basis for R^2.

Working:

Two vectors form a basis for R^2 when the determinant of the matrix with those vectors as columns is not zero.

$$A = \begin{bmatrix}-4 & -2\\-3 & 4\end{bmatrix}$$

$$\det(A) = (-4)(4) - (-2)(-3) = -22.$$

The determinant is not zero, so the vectors are independent and span R^2.

Teacher answers & working

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EA1

Express the given vector as a linear combination of the two vectors.

Express $\begin{bmatrix}-20\\-8\end{bmatrix}$ as a linear combination of $\begin{bmatrix}-2\\1\end{bmatrix}$ and $\begin{bmatrix}-3\\-3\end{bmatrix}$.

Answer:

$\begin{bmatrix}-20\\-8\end{bmatrix} = 4\begin{bmatrix}-2\\1\end{bmatrix} + 4\begin{bmatrix}-3\\-3\end{bmatrix}$

Working:

Let $\begin{bmatrix}-20\\-8\end{bmatrix} = a\begin{bmatrix}-2\\1\end{bmatrix} + b\begin{bmatrix}-3\\-3\end{bmatrix}$.

Equating components gives −2a − 3b = −20 and a − 3b = −8.

Solving the two simultaneous equations gives a = 4 and b = 4.

$\begin{bmatrix}-20\\-8\end{bmatrix} = 4\begin{bmatrix}-2\\1\end{bmatrix} + 4\begin{bmatrix}-3\\-3\end{bmatrix}$.

EA2

Decide whether the vector belongs to the span of the given vectors.

Decide whether $\begin{bmatrix}-8\\-3\end{bmatrix}$ is in span$\{\begin{bmatrix}-3\\-1\end{bmatrix}, \begin{bmatrix}-9\\-3\end{bmatrix}\}$.

Answer:

No, the vector is not in the span.

Working:

$\begin{bmatrix}-9\\-3\end{bmatrix} = 3\begin{bmatrix}-3\\-1\end{bmatrix}$, so the span is the line containing scalar multiples of $\begin{bmatrix}-3\\-1\end{bmatrix}$.

For $\begin{bmatrix}-8\\-3\end{bmatrix}$ to be in the span, the component ratios must match.

The ratios do not match, so no scalar t gives $\begin{bmatrix}-8\\-3\end{bmatrix} = t\begin{bmatrix}-3\\-1\end{bmatrix}$.

Therefore $\begin{bmatrix}-8\\-3\end{bmatrix}$ is not in the span.

Teacher answers & working

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EB1

Determine whether the three vectors are linearly independent.

Determine whether $\begin{bmatrix}1\\0\\2\end{bmatrix}$, $\begin{bmatrix}0\\1\\0\end{bmatrix}$, and $\begin{bmatrix}0\\0\\3\end{bmatrix}$ are linearly independent.

Answer:

Linearly independent.

Working:

Put the vectors as columns of a matrix.

$$\begin{bmatrix}1&0&0\\0&1&0\\2&0&3\end{bmatrix}$$

This matrix is triangular after reading the first two rows, and the third pivot is 3.

There is a pivot in each column, so the only solution to $a\begin{bmatrix}1\\0\\2\end{bmatrix} + b\begin{bmatrix}0\\1\\0\end{bmatrix} + c\begin{bmatrix}0\\0\\3\end{bmatrix} = \begin{bmatrix}0\\0\\0\end{bmatrix}$ is a = b = c = 0.

Therefore the vectors are linearly independent.